

Active thermal management relies on cooling technologies using a substance such as air, liquid and refrigerant cooling. It involves an external device that helps dissipate heat. Active methods are generally more expensive and complex than passive techniques.

Passive methods rely upon the thermodynamics of heat transfer conduction, convection and radiation. Passive battery cooling includes metal heat-sinks, PCMs and specialised heat-shields. These are typically less expensive than active technologies and are easier to put in place.

Development in battery manufacturing

Lithium batteries that use cobalt are prone to thermal runaway and fire and, hence, require additional equipment for thermal monitoring and cooling. **Lithium battery chemistries that do not contain cobalt, such as lithium iron phosphate, are not prone to thermal runaway or fire and, are, hence, quite safe.**

Scientists at Massachusetts Institute of Technology (MIT), Stanford University and Toyota Research Institute (TRI) discovered that combining comprehensive experimental data and artificial intelligence (AI) can predict the useful life of lithium-ion batteries before their capacities start to wane. Their machine learning model enables the algorithm to predict how many more cycles each battery would last, based on voltage declines and a few other factors among early cycles.

Lithium-sulphur batteries offer a theoretical energy density more than five times that of lithium-ion batteries. Researchers at Chalmers University of Technology, Sweden, have developed a porous, sponge-like aerogel, made of reduced graphene oxide, which acts as a free-standing electrode in the battery cell and allows for better and higher utilisation of sulphur.

Dr Suzzi adds, "During battery pack development at AVL, thermal runaway is modelled applying CAE methods, allowing time-effective and preventive evaluation of design countermeasures. Different design scenarios are virtually evaluated, for

example, specific cell and module geometry, electrical connections, thermally-resistant layers serving as heat barriers, different cooling concepts, different triggering methods, position of venting plugs/burst disks on the battery housing and so on. These investigations are of significant relevance for developing strategies to prevent or postpone thermal propagation, as well as to meet safety requirements for batteries in EVs."

Why venting is needed

Jake Sanders, product development manager, Donaldson Venting Solutions, says, "Unlike smaller lithium-ion batteries used in home electronics, automotive lithium-ion batteries need robust protection from harsh external conditions, along with adequate venting for temperature and pressure fluctuations."

A battery pack for an EV may have hundreds of cells contained within several modules. In addition to keeping out contaminants, battery enclosures need to handle pressure differentials between the enclosure interior and the surrounding atmosphere. Pressure differential can fluctuate widely during normal vehicle operation due to ambient temperature changes, heat generation in cells and atmospheric pressure changes. To prevent the collapse or compromise of the protective shell, enclosures must be vented to equalise internal and external pressures, while releasing gases produced in a rapid pressure. **Whether gradual or sudden, pressure buildup can stress seals, leading to leaks and potential explosions.**

To sum up

Lithium-ion batteries can be a safe and reliable means of energy for EVs by means of proper cell selection and design and manufacturing, including an advanced battery management system to prevent thermal runaway. Maintenance-free batteries do not mean that these will operate perfectly forever! Hence, it is important to have the batteries checked at least once a year. Without proper maintenance, monitoring and proper installation, the batteries will end up costing a lot in the end. **EFY**

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